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Switching regulator and control method thereof

Abstract

A switching regulator includes: a power stage circuit, a control circuit and an operation clock signal generator circuit. The operation clock signal generator circuit includes: a time point option unit generating a time point option signal according to a phase node voltage during a ringing period subsequent to a blanking period, to indicate at least one available turn-on time point, or generating a lowest voltage time point signal according to the phase node voltage during a tolerance period, to indicate a lowest voltage time point; and a time point deciding unit deciding the tolerance period according to a base clock signal and a tolerable frequency range and select the available turn-on time point or the lowest voltage time point within the tolerance period as a decided time point, to generate the operation clock signal.

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Background/Summary

CROSS REFERENCE

(1) The present invention claims priority to TW 111144678 filed on Nov. 22, 2022.

BACKGROUND OF THE INVENTION

Field of Invention

(2) The present invention relates to a switching regulator and a control method thereof; particularly, it relates to such a switching regulator capable of operating substantially in constant frequency and having an improved light load efficiency, and a control method thereof.

Description

DESCRIPTION OF RELATED ART

(1) FIG. 1 is a diagram showing the light load efficiency and phase node voltage when the lower gate switch is ON in a conventional switching regulator. As shown in FIG. 1, because this conventional switching regulator adopts a single constant switching frequency and because this conventional switching regulator is unable to achieve zero voltage switching (ZVS), the phase node voltage when the lower gate switch is ON is sometimes low (e.g., at a valley) and is sometimes high (e.g., at a peak). As a result, the prior art shown in FIG. 1 has an unstable light load efficiency; the light load efficiency is better only when the phase node voltage is close to the valley.

(2) In another prior art, the time point when the lower gate switch is turned ON is adjusted by taking zero voltage switching into consideration, so as to control the time point when the lower gate switch is turned ON to be at a time point when the phase node voltage of the lower gate switch

is zero to achieve zero voltage switching, thereby improving the power conversion efficiency. Nevertheless, this involves adjusting operation frequency and the operation frequency may be beyond a range acceptable by a load circuit which is required to operate in constant frequency. That is, when a load circuit is required to operate in a specific range (e.g., an LED driver which is applied in a touch screen with a stylus), zero voltage switching cannot be smoothly achieved. A relevant prior art is U.S. Pat. No. 8,629,660B2.

(3) In view of the above, to overcome the drawbacks in the prior art, the present invention proposes a switching regulator which is capable of operating substantially in constant frequency and having an improved light load efficiency, and a control method thereof.

SUMMARY OF THE INVENTION

(4) From one perspective, the present invention provides a switching regulator, comprising: a power stage circuit, which is configured to operably control at least one power switch according to a pulse width modulation (PWM) signal, to switch an inductor coupled to a phase node, thus converting an input voltage to an output voltage; a control circuit, which is configured to operably generate the PWM signal according to an operation clock signal; and an operation clock signal generator circuit, which is configured to operably generate the operation clock signal, wherein the operation clock signal generator circuit includes: a time point option unit, which is configured to operably generate a time point option signal according to a phase node voltage of the phase node during a ringing period subsequent to a blanking period, to indicate at least one available turn-on time point, or which is configured to operably generate a lowest voltage time point signal according to the phase node voltage during a tolerance period, to indicate a lowest voltage time point; and a time point deciding unit, which is configured to operably decide the tolerance period according to a base clock signal and a tolerable frequency range and which is configured to operably select the available turn-on time point or the lowest voltage time point which occurs within the tolerance period to be a decided time point, so as to generate the operation clock signal; wherein the decided time point is a time point which corresponds to a lowest phase node voltage within the tolerance period or an ending time point of the tolerance period; wherein a beginning time point of the tolerance period is a time point obtained by counting a shortest tolerance period from a previous decided time point, wherein the shortest tolerance period corresponds to a highest frequency of the tolerable frequency range; wherein an ending time point of the tolerance period is a time point obtained by counting a longest tolerance period from a previous decided time point, wherein the longest tolerance period corresponds to a lowest frequency of the tolerable frequency range; wherein the ringing period occurs during a dead time when the switching regulator operates in a discontinuous conduction mode (DCM).

(5) In one embodiment, herein the operation clock signal generator circuit further includes: a blanking unit, which is configured to operably generate a blanking signal for defining the blanking period, wherein the blanking unit is configured to operably decide a beginning time point of the blanking period according to the PWM signal; wherein an ending time point of the blanking period is decided by the blanking unit according to the phase node voltage, or the ending time point of the blanking period is decided by selecting a time point prior to the beginning time point of the tolerance period.

(6) In one embodiment, the available turn-on time point is correlated with a valley time point of a ringing signal of the phase node voltage during the ringing period.

(7) In one embodiment, the power stage circuit includes a boost power stage circuit, wherein the output voltage is not lower than 2-fold of the input voltage.

(8) In one embodiment, the ringing period is correlated with an inductance of the inductor and an equivalent capacitance at the phase node.

(9) In one embodiment, the blanking unit includes: a comparison circuit, which is configured to operably compare the phase node voltage with a first voltage threshold, to generate a comparison result; and a logic circuit, which is configured to operably decide the ending time point of the

blanking period according to the comparison result and which is configured to operably decide the beginning time point of the blanking period according to the PWM signal, so as to generate the blanking signal.

(10) In one embodiment, the first voltage threshold includes: a common mode voltage or a knee voltage of the phase node voltage.

(11) In one embodiment, in a case when the time point option unit is configured to generate the time point option signal according to the phase node voltage of the phase node during the ringing period subsequent to the blanking period, the time point option unit includes: a time point searching circuit, which is configured to operably generate a time point searching result according to the blanking signal and the phase node voltage, wherein the time point searching result indicates an occurrence time point of at least one valley or at least one peak of the phase node voltage within the ringing period; and a time point determination circuit, which is configured to operably decide whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is the available turn-on time point, so as to decide at least one available turn-on time point, thereby generating the time point option signal; wherein the time point searching circuit is disabled at the beginning time point of the blanking signal, whereas, the time point searching circuit is enabled at the ending time point of the blanking signal.

(12) In one embodiment, the time point searching circuit includes: a differentiator circuit, which is configured to operably execute a differentiation operation on the phase node voltage, to generate a differentiation signal; and a comparison circuit, which is configured to operably compare the differentiation signal with a common mode signal of the differentiation signal, to generate the time point searching result.

(13) In one embodiment, the time point searching circuit includes: an integrator circuit, which is configured to operably execute an integration operation on the phase node voltage, to generate an integration signal; and a comparison circuit, which is configured to operably compare the integration signal with a common mode signal of the integration signal, to generate the time point searching result.

(14) In one embodiment, the time point searching circuit includes: a current sensing circuit, which is configured to operably sense a current flowing through the phase node, to generate a current signal; and a comparison circuit, which is configured to operably compare the current signal with a common mode signal of the current signal, to generate the time point searching result.

(15) In one embodiment, the time point determination circuit is configured to operably decide whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is the available turn-on time point according to whether the phase node voltage corresponding to each of the occurrence time point is lower than a second voltage threshold.

(16) In one embodiment, the second voltage threshold includes: a common mode voltage of the phase node voltage.

(17) In one embodiment, the time point determination circuit is configured to operably count a sequence order of each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result, so as to decide whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is the available turn-on time point.

(18) In one embodiment, when the sequence order of the occurrence time point of the at least one valley or the at least one peak is an odd, the time point determination circuit decides this occurrence time point of the at least one valley or the at least one peak is the available turn-on time point.

(19) In one embodiment, the time point deciding unit includes: a frequency divider, which is configured to operably generate a high frequency clock signal and a low frequency clock signal according to the base clock signal, the tolerable frequency range and the PWM signal; and a

tolerance period generation circuit, which is configured to operably generate a tolerance period signal according to the high frequency clock signal and the PWM signal, wherein the tolerance period signal is indicative of the tolerance period; wherein when the time point option signal and the tolerance period signal are both at an enable level, the operation clock signal is generated, or otherwise, the operation clock signal is generated at a rising time point of the low frequency clock signal.

(20) In one embodiment, the time point deciding unit further includes: an edge detection circuit, which is configured to operably detect a rising edge of the low frequency clock signal, to decide the rising time point of the low frequency clock signal.

(21) In one embodiment, the power stage circuit includes: a buck power stage circuit, a boost power stage circuit, a buck-boost power stage circuit or a flyback power stage circuit.

(22) In one embodiment, the common mode voltage of the phase node voltage includes: the input voltage.

(23) In one embodiment, the common signal includes: the input voltage.

(24) In one embodiment, when the time point option unit is configured to operably generate the lowest voltage time point signal during the tolerance period, the time point option unit includes: a sample-and-hold circuit, which is enabled by a tolerance period signal and which is configured to operably sample and hold a plurality of the phase node voltages during the tolerance period, to generate the time point option signal; and a lowest value selector coupled to the sample-and-hold circuit, wherein the lowest value selector is configured to operably select a lowest phase node voltage among the plurality of the phase node voltages, to generate the lowest voltage time point signal, wherein the lowest voltage time point signal is indicative of the lowest voltage time point.

(25) In one embodiment, the time point deciding unit includes: a frequency divider, which is configured to operably generate a high frequency clock signal and a low frequency clock signal according to the base clock signal, the tolerable frequency range and the PWM signal; a tolerance period generation circuit, which is configured to operably generate a tolerance period signal according to the high frequency clock signal and the PWM signal, wherein the tolerance period signal is indicative of the tolerance period; and a finite-state machine (FSM), which is configured to operably generate the operation clock signal according to the lowest voltage time point signal and the low frequency clock signal.

(26) From another perspective, the present invention provides a control method for controlling a switching regulator, wherein the switching regulator includes: a power stage circuit, which is configured to operably control at least one power switch according to a pulse width modulation (PWM) signal, to switch an inductor coupled to a phase node, thus converting an input voltage to an output voltage; the control method comprising following steps: generating a time point option signal according to a phase node voltage of the phase node during a ringing period subsequent to a blanking period, to indicate at least one available turn-on time point, or generating a lowest voltage time point signal according to the phase node voltage during a tolerance period, to indicate a lowest voltage time point; and deciding the tolerance period according to a base clock signal and a tolerable frequency range and selecting the available turn-on time point or the lowest voltage time point which occurs within the tolerance period to be a decided time point, so as to generate the operation clock signal; wherein the decided time point is a time point which corresponds to a lowest phase node voltage within the tolerance period or an ending time point of the tolerance period; wherein a beginning time point of the tolerance period is a time point obtained by counting a shortest tolerance period from a previous decided time point, wherein the shortest tolerance period corresponds to a highest frequency of the tolerable frequency range; wherein an ending time point of the tolerance period is a time point obtained by counting a longest tolerance period from a previous decided time point, wherein the longest tolerance period corresponds to a lowest frequency of the tolerable frequency range.

(27) In one embodiment, the control method further comprises: generating a blanking signal which

defines the blanking period; wherein a beginning time point of the blanking period is decided according to the PWM signal, and an ending time point of the blanking period is decided according to the phase node voltage, or the ending time point of the blanking period is decided by selecting a time point prior to the beginning time point of the tolerance period.

(28) In one embodiment, the step for generating the blanking signal includes: comparing the phase node voltage with a first voltage threshold, to generate a comparison result; and deciding the ending time point of the blanking period according to the comparison result and deciding the beginning time point of the blanking period according to the PWM signal, so as to generate the blanking signal.

(29) In one embodiment, the step for generating the time point option signal according to the phase node voltage of the phase node during the ringing period subsequent to the blanking period includes: generating a time point searching result according to the blanking signal and the phase node voltage, wherein the time point searching result indicates an occurrence time point of at least one valley or at least one peak of the phase node voltage within the ringing period; and determining whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is the available turn-on time point, to decide at least one available turn-on time point, thereby generating the time point option signal.

(30) In one embodiment, the step for generating the time point searching result according to the blanking signal and the phase node voltage includes: executing a differentiation operation on the phase node voltage, to generate a differentiation signal; and comparing the differentiation signal with a common mode signal of the differentiation signal, to generate the time point searching result.

(31) In one embodiment, the step for generating the time point searching result according to the blanking signal and the phase node voltage includes: executing an integration operation on the phase node voltage, to generate an integration signal; and comparing the integration signal with a common mode signal of the integration signal, to generate the time point searching result.

(32) In one embodiment, the step for generating the time point searching result according to the blanking signal and the phase node voltage includes: sensing a current flowing through the phase node, to generate a current signal; and comparing the current signal with a common mode signal of the current signal, to generate the time point searching result.

(33) In one embodiment, the step for determining whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is the available turn-on time point is determined according to whether the phase node voltage corresponding to each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is lower than a second voltage threshold.

(34) In one embodiment, the step for determining whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is the available turn-on time point includes: counting a sequence order of each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result.

(35) In one embodiment, the step for deciding the tolerance period according to the base clock signal and the tolerable frequency range and selecting the available turn-on time point or the lowest voltage time point which occurs within the tolerance period to be the decided time point to generate the operation clock signal includes: generating a high frequency clock signal and a low frequency clock signal according to the base clock signal, the tolerable frequency range and the PWM signal; and generating a tolerance period signal according to the high frequency clock signal and the low frequency clock signal, wherein the tolerance period signal is indicative of the tolerance period; wherein when the time point option signal and the tolerance period signal are both at an enable level, the operation clock signal is generated, or otherwise, the operation clock signal is generated at a rising time point of the low frequency clock signal.

(36) In one embodiment, the step for deciding the tolerance period according to the base clock

signal and the tolerable frequency range and selecting the available turn-on time point or the lowest voltage time point which occurs within the tolerance period to be the decided time point to generate the operation clock signal includes: detecting a rising edge of the low frequency clock signal, to decide the rising time point of the low frequency clock signal.

(37) In one embodiment, the step for generating the lowest voltage time point signal according to the phase node voltage during the tolerance period includes: when being enabled by a tolerance period signal, sampling and holding a plurality of the phase node voltages during the tolerance period, to generate the time point option signal; and selecting a lowest phase node voltage among the plurality of the phase node voltages, to generate the lowest voltage time point signal, wherein the lowest voltage time point signal is indicative of the lowest voltage time point.

(38) In one embodiment, the step for deciding the tolerance period according to the base clock signal and the tolerable frequency range and selecting the available turn-on time point or the lowest voltage time point which occurs within the tolerance period to be the decided time point to generate the operation clock signal includes: generating a high frequency clock signal and a low frequency clock signal according to the base clock signal, the tolerable frequency range and the PWM signal; generating a tolerance period signal according to the high frequency clock signal and the low frequency clock signal, wherein the tolerance period signal is indicative of the tolerance period; and generating the operation clock signal according to the lowest voltage time point signal and the low frequency clock signal.

(39) Advantages of the present invention include: that the present invention can operate substantially in constant frequency to a certain degree, and that the present invention has an improved light load efficiency.

(40) The objectives, technical details, features, and effects of the present invention will be better understood with regard to the detailed description of the embodiments below, with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

(41) FIG. 1 is a diagram showing the light load efficiency and phase node voltage when the lower gate switch is ON in a conventional switching regulator.

(42) FIG. 2A shows a schematic circuit block diagram of a switching regulator according to an embodiment of the present invention.

(43) FIG. 2B shows a schematic circuit block diagram of a switching regulator according to another embodiment of the present invention.

(44) FIG. 2C shows a schematic circuit block diagram of a switching regulator according to yet another embodiment of the present invention.

(45) FIG. 2D shows a schematic circuit block diagram of a switching regulator according to still another embodiment of the present invention.

(46) FIG. 3A to FIG. 3K show different types of power stage circuits that the present invention can be applied to.

(47) FIG. 4 shows a schematic circuit diagram of a blanking unit in a switching regulator according to an embodiment of the present invention.

(48) FIG. 5 shows a schematic block diagram of a time point option unit in a switching regulator according to an embodiment of the present invention.

(49) FIG. 6A shows a schematic circuit block diagram of a time point option unit in a switching regulator according to an embodiment of the present invention.

(50) FIG. 6B illustrates a signal waveform diagram depicting relevant signals within a time point option unit of FIG. 6A according to an embodiment of the present invention.

(51) FIG. 7A shows a schematic circuit block diagram of a time point option unit in a switching regulator according to another embodiment of the present invention.

(52) FIG. 7B illustrates a signal waveform diagram depicting relevant signals within a time point option unit of FIG. 7A according to an embodiment of the present invention.

- (53) FIG. 8 shows a schematic circuit block diagram of a time point option unit in a switching regulator according to yet another embodiment of the present invention.
- (54) FIG. 9 shows a schematic circuit diagram of a time point option unit in a switching regulator according to still another embodiment of the present invention.
- (55) FIG. 10 shows a schematic circuit diagram of a time point option unit in a switching regulator according to still another embodiment of the present invention.
- (56) FIG. 11 shows a schematic circuit block diagram of a time point option unit in a switching regulator according to still another embodiment of the present invention.
- (57) FIG. 12 shows a schematic circuit block diagram of a time point deciding unit in a switching regulator according to an embodiment of the present invention.
- (58) FIG. 13 shows a schematic circuit block diagram of a time point deciding unit in a switching regulator according to another embodiment of the present invention.
- (59) FIG. 14 illustrates a signal waveform diagram depicting relevant signals in a switching regulator according to an embodiment of the present invention.
- (60) FIG. 15 illustrates a signal waveform diagram depicting relevant signals in a switching regulator according to another embodiment of the present invention.
- (61) FIG. 16 is a schematic diagram showing the light load efficiency versus load current of a switching regulator of the present invention and the light load efficiency versus load current of a conventional switching regulator.
- (62) FIG. 17 to FIG. 29 show flow chart diagrams of a control method of a switching regulator according to an embodiment of the present invention.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(63) The drawings as referred to throughout the description of the present invention are for illustration only, to show the interrelations between the circuits and the signal waveforms, but not drawn according to actual scale of circuit sizes and signal amplitudes and frequencies. For better understanding the essence of the present invention, practical implementation details will be described in the embodiments below. It should be understood that such details are not for limiting the broadest scope of the present invention.

(64) Please refer to FIG. 2A, which shows a schematic circuit block diagram of a switching regulator according to an embodiment of the present invention. As shown in FIG. 2A, the switching regulator **20** of the present invention includes: a power stage circuit **201**, a control circuit **202** and an operation clock signal generator circuit **203**. The power stage circuit **201** is configured to operably switch at least one power switch therein (which will be shown later) according to a pulse width modulation (PWM) signal $Spwm$, to switch an inductor (which will be shown later) coupled to a phase node LX, thus converting an input voltage V_{in} to an output voltage V_{out} , wherein the phase node LX has a phase node voltage V_{lx} . In one embodiment, the aforementioned output voltage V_{out} is provided to a load circuit which requires to operate under a substantial constant frequency of the PWM signal $Spwm$ (i.e., the operation frequency of the power switch), wherein such load circuit can be, for example but not limited to, a touch screen with a stylus. The control circuit **202** is configured to operably generate the PWM signal $Spwm$ according to an operation clock signal CLK_OP. The operation clock signal generator circuit **203** is configured to operably generate the operation clock signal CLK_OP.

(65) Please refer to FIG. 2A in conjunction with FIG. 14. FIG. 14 illustrates a signal waveform diagram depicting relevant signals in a switching regulator according to an embodiment of the present invention. The operation clock signal generator circuit **203** includes: a time point option unit **2031** and a time point deciding unit **2032**. The time point option unit **2031** is configured to operably generate a lowest voltage time point signal TP_Vmin according to the phase node voltage V_{lx} during a tolerance period T_{pp} , to indicate a lowest voltage time point at which the phase node voltage V_{lx} is at its lowest voltage during the tolerance period T_{pp} . The time point deciding unit **2032** is configured to operably decide the tolerance period T_{pp} according to a base clock signal

CLK_B and a tolerable frequency range, and is configured to operably select the lowest voltage time point to be a decided time point according to the lowest voltage time point signal TP_Vmin, to generate the operation clock signal CLK_OP.

(66) The decided time point is a time point which corresponds to when the phase node voltage V_{lx} is lowest within the tolerance period T_{pp} . The beginning time point of the tolerance period T_{pp} is a time point obtained by counting a shortest tolerance period T_{cmin} from a previous decided time point, wherein the shortest tolerance period T_{cmin} corresponds to a highest frequency of the tolerable frequency range. The ending time point of the tolerance period T_{pp} is a time point obtained by counting a longest tolerance period T_{cmax} from a previous decided time point, wherein the longest tolerance period T_{cmax} corresponds to a lowest frequency of the tolerable frequency range. It is worthwhile noting that, there is a ringing period T_{rin} occurring during a dead time when the switching regulator **20** operates in a discontinuous conduction mode (DCM). When the switching regulator operates in DCM, when a load circuit powered by the output voltage V_{out} is in a light load condition and when the power switches in the power stage circuit **201** are all OFF, the phase node voltage V_{lx} will generate a ringing signal having a sinusoidal waveform. The ringing period T_{rin} of the ringing signal is correlated with an inductance of the aforementioned inductor and an equivalent capacitance at the phase node LX.

(67) Please refer to FIG. 2B, which shows a schematic circuit block diagram of a switching regulator according to another embodiment of the present invention. As shown in FIG. 2B, the switching regulator **20** of the present invention includes: a power stage circuit **201**, a control circuit **202** and an operation clock signal generator circuit **203**. The power stage circuit **201** is configured to operably switch at least one power switch (which will be shown later) according to a pulse width modulation (PWM) signal $Spwm$, to switch an inductor (which will be shown later) coupled to a phase node LX, thus converting an input voltage V_{in} to an output voltage V_{out} , wherein the phase node LX has a phase node voltage V_{lx} . In one embodiment, the aforementioned output voltage V_{out} is provided to a load circuit which requires to operate under a substantial constant frequency of the PWM signal $Spwm$ (i.e., the operation frequency of the power switch), wherein such load circuit can be, for example but not limited to, a touch screen with a stylus. The control circuit **202** is configured to operably generate the PWM signal $Spwm$ according to an operation clock signal CLK_OP. The operation clock signal generator circuit **203** is configured to operably generate the operation clock signal CLK_OP.

(68) Please refer to FIG. 2B in conjunction with FIG. 14. The operation clock signal generator circuit **203** includes: a time point option unit **2031**, a time point deciding unit **2032** and a blanking unit **2033**. The time point option unit **2031** is configured to operably generate a time point option signal TP_CAN according to the phase node voltage V_{lx} of the phase node LX during a ringing period T_{rin} subsequent to a blanking period T_{blk} , to indicate at least one available turn-on time point. The time point deciding unit **2032** is configured to operably decide the tolerance period T_{pp} according to a base clock signal CLK_B and a tolerable frequency range, and is configured to operably select an available turn-on time point occurring within the tolerance period T_{pp} to be a decided time point, to generate the operation clock signal CLK_OP.

(69) In one embodiment, the available turn-on time point is within the tolerance period T_{pp} . Please refer to FIG. 2B along with FIG. 14. The blanking unit **2033** is configured to operably generate a blanking signal BLK which defines the blanking period T_{blk} . In one embodiment, the blanking unit **2033** is configured to operably decide a beginning time point of the blanking period T_{blk} according to the PWM signal $Spwm$. In one embodiment, the ending time point of the blanking period T_{blk} is decided by the blanking unit **2033** according to the phase node voltage V_{lx} , or by selecting a time point before the beginning time point of the tolerance period T_{pp} .

(70) In one embodiment, the decided time point is a time point which corresponds to a lowest the phase node voltage V_{lx} within the tolerance period T_{pp} , or the decided time point is the ending time point of the tolerance period T_{pp} . Please refer to FIG. 14. In one embodiment, the beginning

time point of the tolerance period T_{pp} a time point obtained by counting a shortest tolerance period T_{cmin} from a previous decided time point, wherein the shortest tolerance period T_{cmin} corresponds to a highest frequency of the tolerable frequency range. In one embodiment, the ending time point of the tolerance period T_{pp} is a time point obtaining by counting a longest tolerance period T_{cmax} from a previous decided time point, wherein the longest tolerance period T_{cmax} corresponds to a lowest frequency of the tolerable frequency range. As described above, there is a ringing period T_{rin} occurring during the dead time when the switching regulator **20** operates in a discontinuous conduction mode (DCM).

(71) Please refer to FIG. **14**. In one embodiment, the available turn-on time point is correlated with a valley time point T_v of the ringing signal of the phase node voltage V_{lx} during the ringing period T_{rin} . In one embodiment, the power stage circuit **201** includes a boost power stage circuit, and the output voltage V_{out} of boost power stage circuit is not lower than 2-fold of the input voltage V_{in} of the boost power stage circuit. Please refer to FIG. **14** along with FIG. **2C** and FIG. **2D**. The ringing period T_{rin} of the ringing signal of the phase node voltage V_{lx} is correlated with an inductance of the inductor L and an equivalent capacitance at the phase node LX .

(72) FIG. **2C** shows a schematic circuit block diagram of a switching regulator according to yet another embodiment of the present invention. The power stage circuit **201** of this embodiment is a boost power stage circuit. As shown in FIG. **2C**, in one embodiment, the power stage circuit **201** includes: an inductor L and at least one power switch. The at least one power switch includes: an upper gate switch QH and a lower gate switch QL . The upper gate switch QH is coupled between the output voltage V_{out} and a phase node LX , whereas, the lower gate switch QL is coupled between phase node LX and a ground potential. The inductor L has its one end coupled to the input voltage V_{in} and its another end coupled to the phase node LX . In one embodiment, the upper gate switch QH can be replaced by an upper gate diode. In one embodiment, the lower gate switch QL can be replaced by a lower gate diode.

(73) FIG. **2D** shows a schematic circuit block diagram of a switching regulator according to still another embodiment of the present invention. The switching regulator of this embodiment shown in FIG. **2C** is similar to the switching regulator of the embodiment shown in FIG. **2B**, but is different in that: the upper gate switch of this embodiment is a Schottky diode, whereas, a lower gate switch QL is a Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET).

(74) FIG. **3A** to FIG. **3K** show several embodiments of a switching regulator that the present invention is applicable to. The power stage circuit **201** of FIG. **2A** can be implemented as any one of the power stage circuit topologies shown in FIG. **3A** to FIG. **3K**, wherein FIG. **3A** to FIG. **3K** show synchronous and asynchronous buck, boost, buck-boost, inverting buck-boost and flyback power stage circuits, respectively.

(75) FIG. **4** shows a schematic circuit diagram of a blanking unit in a switching regulator according to an embodiment of the present invention. As shown in FIG. **4**, the blanking unit **2033** includes: a comparison circuit **20331** and a logic circuit **20332**. The comparison circuit **20331** is configured to operably compare the phase node voltage V_{lx} with a first voltage threshold V_{th1} , to generate a comparison result CPO . In one embodiment, the first voltage threshold V_{th1} includes: a common mode voltage V_{lx_cm} or a knee voltage of the phase node voltage V_{lx} . In one embodiment, in a case when the power stage circuit **201** is a boost power stage circuit, the common mode voltage V_{lx_cm} of the phase node voltage V_{lx} includes the input voltage V_{in} . The logic circuit **20332** is configured to operably decide the ending time point of the blanking period T_{blk} according to the comparison result CPO and is configured to operably decide the beginning time point of the blanking period T_{blk} according to the PWM signal $Spwm$, so as to generate the blanking signal BLK . In one embodiment, the logic circuit **20332** includes: an edge detector and a flip-flop.

(76) It is worthwhile noting that, different types of power stage circuits have different types of common mode voltages V_{lx_cm} of the phase node voltage V_{lx} . As exemplified by the embodiment shown in FIG. **4**, in a case when the power stage circuit is a boost power stage circuit, the common

mode voltage V_{lx_cm} of the phase node voltage V_{lx} includes the input voltage V_{in} , whereas, in a case when the power stage circuit is a buck power stage circuit, the common mode voltage V_{lx_cm} of the phase node voltage V_{lx} includes the output voltage V_{out} , whereas, in a case when the power stage circuit is a buck-boost power stage circuit, the common mode voltage V_{lx_cm} of the phase node voltage V_{lx} includes a ground potential. That the common mode voltage V_{lx_cm} includes the “input voltage V_{in} ” or “output voltage V_{out} ” refers to the voltage level of the input voltage V_{in} or output voltage V_{out} in steady state.

(77) FIG. 5 shows a schematic block diagram of a time point option unit in a switching regulator according to an embodiment of the present invention. As shown in FIG. 5 and FIG. 14, in a case when the time point option unit **2031** is configured to operably generate a time point option signal TP_CAN according to a phase node voltage V_{lx} of the phase node LX during a ringing period $Trin$ subsequent to a blanking period $Tblk$, the time point option unit **2031** includes: a time point searching circuit **20311** and a time point determination circuit **20312**. The time point searching circuit **20311** is configured to operably generate a time point searching result TSO according to the blanking signal BLK and the phase node voltage V_{lx} , wherein the time point searching result TSO indicates an occurrence time point of at least one valley or at least one peak of the phase node voltage V_{lx} within the ringing period $Trin$. The time point determination circuit **20312** is configured to operably decide whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result TSO is an available turn-on time point, to decide at least one available turn-on time point, thereby generating the time point option signal TP_CAN .

(78) In one embodiment, the time point determination circuit **20312** is configured to operably decide whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result TSO is the available turn-on time point according to whether the phase node voltage V_{lx} corresponding to each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result TSO is lower than a second voltage threshold. For example, when the phase node voltage V_{lx} corresponding to an occurrence time point of a valley or a peak in the time point searching result TSO is lower than a second voltage threshold (which is, for example but not limited to, the common mode voltage V_{lx_cm} of the phase node voltage V_{lx}), the time point determination circuit **20312** decides that this occurrence time point of the valley or the peak in the time point searching result TSO is an available turn-on time point. In another embodiment, the time point determination circuit **20312** is configured to operably count a sequence order of each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result TSO, so as to decide whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result TSO is an available turn-on time point. For example, when an occurrence time point of a certain valley or a certain peak is an odd numbered valley or an odd numbered peak, the time point determination circuit **20312** decides that the occurrence time point of this valley or this peak is an available turn-on time point. In one embodiment, the time point searching circuit **20311** is disabled at the beginning time point of the blanking signal BLK, whereas, the time point searching circuit **20311** is enabled at the ending time point of the blanking signal BLK.

(79) FIG. 6A shows a schematic circuit block diagram of a time point option unit in a switching regulator according to an embodiment of the present invention. In one embodiment, as shown in FIG. 6A, the time point searching circuit **20311** includes: a differentiator circuit **203111a** and a comparison circuit **203112**. The differentiator circuit **203111a** is configured to operably execute a differentiation operation on the phase node voltage V_{lx} , to generate a differentiation signal V_{dlx} . The comparison circuit **203112** is configured to operably compare the differentiation signal V_{dlx} with a common mode signal V_{dlx_cm} of the differentiation signal V_{dlx} , to generate the time point searching result TSO. The time point determination circuit **20312** of this embodiment shown in FIG. 6A is similar to the time point determination circuit **20312** of the embodiment shown in FIG.

5, so the details thereof are not redundantly explained here.

(80) FIG. 6B illustrates a signal waveform diagram depicting relevant signals within a time point option unit of FIG. 6A according to an embodiment of the present invention. The phase node voltage V_{lx} , the common mode voltage V_{lx_cm} of the phase node voltage V_{lx} , the blanking signal BLK, the differentiation signal V_{dlx} , the common mode signal V_{dlx_cm} of the differentiation signal V_{dlx} and the time point option signal TP_CAN are illustrated in FIG. 6B. As shown in FIG. 6B, at time point t1, time point t2 and time point t3, the differentiation signal V_{dlx} is zero and in the meantime the phase node voltages V_{lx} corresponding to the differentiation signal V_{dlx} at these time points are lower than a second voltage threshold (which is, for example but not limited to, the common mode voltage V_{lx_cm} of the phase node voltage V_{lx}). Or, alternatively, at time point t1, time point t2 and time point t3, the differentiation signal V_{dlx} is at an occurrence time point of a first valley or peak, a third valley or peak, and a fifth valley or peak, respectively (i.e., at time point t1, time point t2 and time point t3, the differentiation signal V_{dlx} is at an occurrence time point of an odd numbered valley or peak, wherein valleys and peaks are both counted). Thus, the time point t1, the time point t2 and the time point t3 are all available turn-on time points.

(81) FIG. 7A shows a schematic circuit block diagram of a time point option unit in a switching regulator according to another embodiment of the present invention. As shown in FIG. 7A, in one embodiment, the time point searching circuit 20311 includes: an integrator circuit 203111b and a comparison circuit 203112. The integrator circuit 203111b is configured to operably execute an integration operation on the phase node voltage V_{lx} , to generate an integration signal V_{intg} . The comparison circuit 203112 is configured to operably compare the integration signal V_{intg} with a common mode signal V_{intg_cm} of the integration signal V_{intg} , to generate the time point searching result TSO. In one embodiment, the common mode signal V_{intg_cm} of the integration signal V_{intg} includes the input voltage V_{in} . The time point determination circuit 20312 of this embodiment shown in FIG. 7A is similar to the time point determination circuit 20312 of the embodiment shown in FIG. 5, so the details thereof are not redundantly explained here.

(82) FIG. 7B illustrates a signal waveform diagram depicting relevant signals within a time point option unit of FIG. 7A according to an embodiment of the present invention. The phase node voltage V_{lx} , the common mode voltage V_{lx_cm} of the phase node voltage V_{lx} , the blanking signal BLK, an integration signal V_{intg} , the common mode signal V_{intg_cm} of the integration signal V_{intg} and the time point option signal TP_CAN are illustrated in FIG. 7B. As shown in FIG. 7B, at time point t1, time point t2 and time point t3, the integration signal V_{intg} is zero and in the meantime the phase node voltages V_{lx} corresponding to the integration signals V_{intg} at these time points are lower than a second voltage threshold (which is, for example but not limited to, the common mode voltage V_{lx_cm} of the phase node voltage V_{lx}). Or, alternatively, at time point t1, time point t2 and time point t3, the integration signals V_{intg} are at an occurrence time point of a first valley or peak, a third valley or peak, and a fifth valley or peak, respectively (i.e., at time point t1, time point t2 and time point t3, the integration signal V_{intg} is at an occurrence time point of an odd numbered valley or peak, wherein valleys and peaks are both counted). Thus, the time point t1, the time point t2 and the time point t3 are all available turn-on time points.

(83) FIG. 8 shows a schematic circuit block diagram of a time point option unit in a switching regulator according to yet another embodiment of the present invention. As shown in FIG. 8, in one embodiment, the time point searching circuit 20311 includes: a current sensing circuit 203111c and a comparison circuit 203112. The current sensing circuit 203111c is configured to operably sense a current flowing through the phase node LX, to generate a current signal V_{cs} . The comparison circuit 203112 is configured to operably compare the current signal V_{cs} with a common mode signal V_{cs_cm} of the current signal V_{cs} , to generate the time point searching result TSO. In one embodiment, the common mode signal V_{cs_cm} of the current signal V_{cs} includes the input voltage V_{in} . The time point determination circuit 20312 of this embodiment shown in FIG. 8 is similar to the time point determination circuit 20312 of the embodiment shown in FIG. 5, so the details

thereof are not redundantly explained here.

(84) FIG. 9 shows a schematic circuit diagram of a time point option unit in a switching regulator according to still another embodiment of the present invention. The comparison circuit **203112** of this embodiment shown in FIG. 9 is similar to the comparison circuit **203112** of the embodiment shown in FIG. 6A, so the details thereof are not redundantly explained here. The differentiator circuit **203111a** of this embodiment shown in FIG. 9 is an exemplary embodiment of the differentiator circuit **203111a** shown in FIG. 6A. As shown in FIG. 9, in one embodiment, the differentiator circuit **203111a** includes: an error amplifier **2031111**, a resistor R_d and a capacitor C_d . The capacitor C_d has one end coupled to a negative input end of the error amplifier **2031111** and has another end coupled to the phase node voltage V_{lx} . The resistor R_d has one end coupled to an output end of the error amplifier **2031111** and has another end coupled to the negative input end of the error amplifier **2031111**. The error amplifier **2031111** has its positive input end coupled to the common mode signal V_{dlx_cm} of the differentiation signal V_{dlx} .

(85) The time point determination circuit **20312** of this embodiment shown in FIG. 9 is an exemplary embodiment of the time point determination circuit **20312** shown in FIG. 6A. In one embodiment, the time point determination circuit **20312** includes: a comparison circuit **203121** and an AND gate **203122**. When the phase node voltage V_{lx} is smaller than the common mode voltage V_{lx_cm} of the phase node voltage V_{lx} and when the time point searching result TSO is at an enable level, the time point option signal TP_CAN is switched to an enable level. When the blanking signal BLK is switched to an enable level, the positive input end of the comparison circuit **203121** is coupled to the output voltage V_{out} via a switch S_d , so as to implement the blanking period T_{blk} .

(86) FIG. 10 shows a schematic circuit diagram of a time point option unit in a switching regulator according to still another embodiment of the present invention. The comparison circuit **203112** of this embodiment shown in FIG. 10 is similar to the comparison circuit **203112** of the embodiment shown in FIG. 7A, so the details thereof are not redundantly explained here. The time point determination circuit **20312** of this embodiment shown in FIG. 10 is similar to the time point determination circuit **20312** of the embodiment shown in FIG. 9, so the details thereof are not redundantly explained here. The integrator circuit **203111b** of this embodiment shown in FIG. 10 is an exemplary embodiment of the integrator circuit **203111b** shown in FIG. 7A. As shown in FIG. 10, in one embodiment, the integrator circuit **203111b** includes: a resistor R_i and a capacitor C_i . The resistor R_i and the capacitor C_i are connected in series between the phase node voltage V_{lx} and the input voltage V_{in} . A node N_i between the resistor R_i and the capacitor C_i is coupled to the positive input end of the comparison circuit **203112**. In one embodiment, the node N_i is coupled to the phase node voltage V_{lx} via a switch S_i or the node N_i is coupled to a voltage higher than phase node voltage V_{lx} . When the blanking signal BLK is switched to an enable level, the node N_i is coupled to the phase node voltage V_{lx} via the switch S_i or the node N_i is coupled to the voltage higher than phase node voltage V_{lx} , so as to implement the blanking period T_{blk} .

(87) FIG. 11 shows a schematic circuit block diagram of a time point option unit in a switching regulator according to still another embodiment of the present invention. Please refer to FIG. 11 along with FIG. 14. In a case when the time point option unit **2031'** is configured to operably generate a lowest voltage time point signal TP_Vmin during the tolerance period T_{pp} , the time point option unit **2031'** includes: a sample-and-hold circuit **203111'**, a lowest value selector **203112'**, a transmission gate **203113'**, an analog-to-digital converter (ADC) **203114'** and a register **203115'**. The sample-and-hold circuit **203111'** is enabled by a tolerance period signal PP and is configured to operably sample and hold the phase node voltage V_{lx} during the tolerance period T_{pp} , to generate an analog time point option signal TP_CANa . The thus generated analog time point option signal TP_CANa is delivered to the ADC **203114'** via the transmission gate **203113'**. The ADC **203114'** performs analog-to-digital conversion on the analog time point option signal TP_CANa , to generate a digital time point option signal TP_CANd . The lowest value selector

203112' is coupled to the sample-and-hold circuit **203111'**, wherein the lowest value selector **203112'** is configured to operably select a lowest phase node voltage V_{lx} among the sampled phase node voltages V_{lx} , to generate the lowest voltage time point signal TP_Vmin which is stored in the register **203115'**, wherein the lowest voltage time point signal TP_Vmin is indicative of the lowest voltage time point.

(88) FIG. 12 shows a schematic circuit block diagram of a time point deciding unit in a switching regulator according to an embodiment of the present invention. As shown in FIG. 12, the time point deciding unit **2032** includes: a frequency divider **20321** and a tolerance period generation circuit **20322**. The frequency divider **20321** is configured to operably generate a high frequency clock signal $HIGH_CLK$ and a low frequency clock signal LOW_CLK according to a base clock signal CLK_B , a tolerable frequency range and the PWM signal $Spwm$. The tolerance period generation circuit **20322** is configured to operably generate a tolerance period signal PP according to the high frequency clock signal $HIGH_CLK$ and a PWM signal $Spwm$, wherein the tolerance period signal PP is indicative of the tolerance period Tpp . When the time point option signal TP_CAN and the tolerance period signal PP are both at an enable level, the operation clock signal CLK_OP is generated, or otherwise, the operation clock signal CLK_OP is generated at a rising time point of the low frequency clock signal LOW_CLK . In one embodiment, the time point deciding unit **2032** further includes: an edge detection circuit **20323**, which is configured to operably detect a rising edge of the low frequency clock signal LOW_CLK , to decide the rising time point of the low frequency clock signal LOW_CLK .

(89) FIG. 13 shows a schematic circuit block diagram of a time point deciding unit in a switching regulator according to another embodiment of the present invention. As shown in FIG. 13, the time point deciding unit **2032'** includes: a frequency divider **20321**, a tolerance period generation circuit **20322** and a finite-state machine (FSM) **20323'**. The frequency divider **20321** is configured to operably generate a high frequency clock signal $HIGH_CLK$ and a low frequency clock signal LOW_CLK according to a base clock signal CLK_B , a tolerable frequency range and a PWM signal $Spwm$. The tolerance period generation circuit **20322** is configured to operably generate a tolerance period signal PP according to the high frequency clock signal $HIGH_CLK$ and the PWM signal $Spwm$, wherein the tolerance period signal PP is indicative of the tolerance period Tpp . The FSM **20323'** is configured to operably generate the operation clock signal CLK_OP according to the lowest voltage time point signal TP_Vmin and the low frequency clock signal LOW_CLK . The time point deciding unit **2032'** of this embodiment is implemented in cooperation with the time point option unit **2031'** of FIG. 11.

(90) FIG. 14 illustrates a signal waveform diagram depicting relevant signals in a switching regulator according to an embodiment of the present invention. The phase node voltage V_{lx} , the common mode voltage V_{lx_cm} of the phase node voltage V_{lx} , the PWM signal $Spwm$, the blanking signal BLK , the time point option signal TP_CAN , the tolerance period signal PP , the base clock signal CLK_B , the operation clock signal CLK_OP and the output voltage V_{out} are illustrated in FIG. 14. As shown in FIG. 14, when a lowest voltage time point or an available turn-on time point falls within the tolerance period Tpp , the operation clock signal CLK_OP is generated at the lowest voltage time point or the available turn-on time point.

(91) FIG. 15 illustrates a signal waveform diagram depicting relevant signals in a switching regulator according to another embodiment of the present invention. The phase node voltage V_{lx} , the common mode voltage V_{lx_cm} of the phase node voltage V_{lx} , the PWM signal $Spwm$, the blanking signal BLK , the time point option signal TP_CAN , the tolerance period signal PP , the base clock signal CLK_B , the operation clock signal CLK_OP and the output voltage V_{out} are illustrated in FIG. 15. As shown in FIG. 15, when neither a lowest voltage time point nor an available turn-on time point falls within the tolerance period Tpp , the operation clock signal CLK_OP is generated at an ending time point of the tolerance period Tpp .

(92) FIG. 16 is a schematic diagram showing the light load efficiency versus load current of a

switching regulator of the present invention and the light load efficiency versus load current of a conventional switching regulator. FIG. 16 demonstrates that, under the same load current, as compared to the prior art, the light load efficiency of the present invention is better.

(93) FIG. 17 to FIG. 29 show flow chart diagrams of a control method of a switching regulator according to an embodiment of the present invention. As shown in FIG. 17, the control method 30 of a switching regulator of the present invention includes: step 301 and step 302. The step 301 includes: generating a time point option signal according to a phase node voltage of the phase node during a ringing period subsequent to a blanking period, to indicate at least one available turn-on time point, or generating a lowest voltage time point signal according to the phase node voltage during a tolerance period, to indicate a lowest voltage time point. Next, the step 302 includes: deciding the tolerance period according to a base clock signal and a tolerable frequency range, and selecting the available turn-on time point or the lowest voltage time point which occurs within the tolerance period as a decided time point, to generate the operation clock signal. As shown in FIG. 18, the control method 30 of the switching regulator of the present invention further includes: step 303. The step 303 includes: generating a blanking signal which defines the blanking period.

(94) In one embodiment, as shown in FIG. 19, the step 303 includes: step 3031 and step 3032. The step 3031 includes: comparing the phase node voltage with a first voltage threshold, to generate a comparison result. Next, the step 3032 includes: deciding the ending time point of the blanking period according to the comparison result and deciding the beginning time point of the blanking period according to the PWM signal, so as to generate the blanking signal.

(95) In one embodiment, as shown in FIG. 20, the step 301 includes: step 3011a and step 3012a. The step 3011a includes: generating a time point searching result according to the blanking signal and the phase node voltage, wherein the time point searching result indicates an occurrence time point of at least one valley or at least one peak of the phase node voltage within the ringing period. Next, the step 3012a includes: determining whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is the available turn-on time point, to decide at least one available turn-on time point, thereby generating the time point option signal.

(96) In one embodiment, as shown in FIG. 21, the step 3011a includes: step 30111a and step 30112a. The step 30111a includes: executing a differentiation operation on the phase node voltage, to generate a differentiation signal. Next, the step 30112a includes: comparing the differentiation signal with a common mode signal of the differentiation signal, to generate the time point searching result.

(97) In another embodiment, as shown in FIG. 22, the step 3011a includes: step 30111b and step 30112b. The step 30111b includes: executing an integration operation on the phase node voltage, to generate an integration signal. Next, the step 30112b includes: comparing the integration signal with a common mode signal of the integration signal, to generate the time point searching result.

(98) In yet another embodiment, as shown in FIG. 23, the step 3011a includes: step 30111c and step 30112c. The step 30111c includes: sensing a current flowing through the phase node, to generate a current signal. Next, the step 30112c includes: comparing the current signal with a common mode signal of the current signal, to generate the time point searching result.

(99) In one embodiment, as shown in FIG. 24, the step 3012a includes: step 30121a. The step 30121a includes: determining whether the phase node voltage corresponding to each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is lower than a second voltage threshold. In another embodiment, as shown in FIG. 25, the step 3012a includes: step 30121b. The step 30121b includes: counting a sequence order of each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result.

(100) In one embodiment, as shown in FIG. 26, the step 302 includes: step 3021a, step 3022a and step 3023a. The step 3021a includes: generating a high frequency clock signal and a low frequency

clock signal according to the base clock signal, the tolerable frequency range and the PWM signal. Next, the step **3022a** includes: generating a tolerance period signal according to the high frequency clock signal and the PWM signal, wherein the tolerance period signal is indicative of the tolerance period. Next, the step **3023a** includes: when the time point option signal and the tolerance period signal are both at an enable level, the operation clock signal is generated, or otherwise, the operation clock signal is generated at a rising time point of the low frequency clock signal. As shown in FIG. 27, in one embodiment, the step **302** further includes: step **3024a**. The step **3024a** includes: detecting a rising edge of the low frequency clock signal, to decide the rising time point of the low frequency clock signal.

(101) In another embodiment, as shown in FIG. 28, the step **301'** includes: step **3011b** and step **3012b**. The step **3011b** includes: when enabled by a tolerance period signal, sampling and holding a plurality of the phase node voltages during the tolerance period, to generate the time point option signal. Next, the step **3012b** includes: selecting a lowest phase node voltage among the plurality of the phase node voltages, to generate the lowest voltage time point signal, wherein the lowest voltage time point signal is indicative of the lowest voltage time point.

(102) In another embodiment, as shown in FIG. 29, the step **302'** includes: step **3021b**, step **3022b** and step **3023b**. The step **3021b** includes: generating a high frequency clock signal and a low frequency clock signal according to the base clock signal, the tolerable frequency range and the PWM signal. Next, the step **3022b** includes: generating a tolerance period signal according to the high frequency clock signal and the low frequency clock signal, wherein the tolerance period signal is indicative of the tolerance period. Next, the step **3023b** includes: generating the operation clock signal according to the lowest voltage time point signal and the low frequency clock signal.

(103) In light of above, on one hand, when a lowest voltage time point or an available turn-on time point falls within the tolerance period, the present invention can adopt the lowest voltage time point or the available turn-on time point as the operation clock signal generation time point; on the other hand, when neither a lowest voltage time point nor an available turn-on time point falls within the tolerance period, the present invention can adopt an ending time point of the tolerance period as the operation clock signal generation time point. By the above-mentioned approach, the present invention can maintain the switching frequency within the tolerance frequency range, and also provide an improved light load efficiency, while keeping the switching regulator to operate by a substantial constant frequency to a certain degree.

(104) The present invention has been described in considerable detail with reference to certain preferred embodiments thereof. It should be understood that the description is for illustrative purpose, not for limiting the broadest scope of the present invention. An embodiment or a claim of the present invention does not need to achieve all the objectives or advantages of the present invention. The title and abstract are provided for assisting searches but not for limiting the scope of the present invention. Those skilled in this art can readily conceive variations and modifications within the spirit of the present invention. For example, to perform an action "according to" a certain signal as described in the context of the present invention is not limited to performing an action strictly according to the signal itself, but can be performing an action according to a converted form or a scaled-up or down form of the signal, i.e., the signal can be processed by a voltage-to-current conversion, a current-to-voltage conversion, and/or a ratio conversion, etc. before an action is performed. It is not limited for each of the embodiments described hereinbefore to be used alone; under the spirit of the present invention, two or more of the embodiments described hereinbefore can be used in combination. For example, two or more of the embodiments can be used together, or, a part of one embodiment can be used to replace a corresponding part of another embodiment. In view of the foregoing, the spirit of the present invention should cover all such and other modifications and variations, which should be interpreted to fall within the scope of the following claims and their equivalents.

Claims

1. A switching regulator, comprising: a power stage circuit, which is configured to operably control at least one power switch according to a pulse width modulation (PWM) signal, to switch an inductor coupled to a phase node, thus converting an input voltage to an output voltage; a control circuit, which is configured to operably generate the PWM signal according to an operation clock signal; and an operation clock signal generator circuit, which is configured to operably generate the operation clock signal, wherein the operation clock signal generator circuit includes: a time point option unit, which is configured to operably generate a time point option signal according to a phase node voltage of the phase node during a ringing period subsequent to a blanking period, to indicate at least one available turn-on time point, or which is configured to operably generate a lowest voltage time point signal according to the phase node voltage during a tolerance period, to indicate a lowest voltage time point; and a time point deciding unit, which is configured to operably decide the tolerance period according to a base clock signal and a tolerable frequency range and which is configured to operably select the available turn-on time point or the lowest voltage time point which occurs within the tolerance period to be a decided time point, so as to generate the operation clock signal; wherein the decided time point is a time point which corresponds to a lowest phase node voltage within the tolerance period or an ending time point of the tolerance period; wherein a beginning time point of the tolerance period is a time point obtained by counting a shortest tolerance period from a previous decided time point, wherein the shortest tolerance period corresponds to a highest frequency of the tolerable frequency range; wherein an ending time point of the tolerance period is a time point obtained by counting a longest tolerance period from a previous decided time point, wherein the longest tolerance period corresponds to a lowest frequency of the tolerable frequency range; wherein the ringing period occurs during a dead time when the switching regulator operates in a discontinuous conduction mode (DCM).
2. The switching regulator of claim 1, wherein the operation clock signal generator circuit further includes: a blanking unit, which is configured to operably generate a blanking signal for defining the blanking period, wherein the blanking unit is configured to operably decide a beginning time point of the blanking period according to the PWM signal; wherein an ending time point of the blanking period is decided by the blanking unit according to the phase node voltage, or the ending time point of the blanking period is decided by selecting a time point prior to the beginning time point of the tolerance period.
3. The switching regulator of claim 1, wherein the available turn-on time point is correlated with a valley time point of a ringing signal of the phase node voltage during the ringing period.
4. The switching regulator of claim 2, wherein the power stage circuit includes a boost power stage circuit, and wherein the output voltage is not lower than 2-fold of the input voltage.
5. The switching regulator of claim 1, wherein the ringing period is correlated with an inductance of the inductor and an equivalent capacitance at the phase node.
6. The switching regulator of claim 2, wherein the blanking unit includes: a comparison circuit, which is configured to operably compare the phase node voltage with a first voltage threshold, to generate a comparison result; and a logic circuit, which is configured to operably decide the ending time point of the blanking period according to the comparison result and which is configured to operably decide the beginning time point of the blanking period according to the PWM signal, so as to generate the blanking signal.
7. The switching regulator of claim 6, wherein the first voltage threshold includes: a common mode voltage or a knee voltage of the phase node voltage.
8. The switching regulator of claim 2, wherein in a case when the time point option unit is configured to generate the time point option signal according to the phase node voltage of the phase node during the ringing period subsequent to the blanking period, the time point option unit

includes: a time point searching circuit, which is configured to operably generate a time point searching result according to the blanking signal and the phase node voltage, wherein the time point searching result indicates an occurrence time point of at least one valley or at least one peak of the phase node voltage within the ringing period; and a time point determination circuit, which is configured to operably decide whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is the available turn-on time point, so as to decide at least one available turn-on time point, thereby generating the time point option signal; wherein the time point searching circuit is disabled at the beginning time point of the blanking signal, whereas, the time point searching circuit is enabled at the ending time point of the blanking signal.

9. The switching regulator of claim 8, wherein the time point searching circuit includes: a differentiator circuit, which is configured to operably execute a differentiation operation on the phase node voltage, to generate a differentiation signal; and a comparison circuit, which is configured to operably compare the differentiation signal with a common mode signal of the differentiation signal, to generate the time point searching result.

10. The switching regulator of claim 8, wherein the time point searching circuit includes: an integrator circuit, which is configured to operably execute an integration operation on the phase node voltage, to generate an integration signal; and a comparison circuit, which is configured to operably compare the integration signal with a common mode signal of the integration signal, to generate the time point searching result.

11. The switching regulator of claim 8, wherein the time point searching circuit includes: a current sensing circuit, which is configured to operably sense a current flowing through the phase node, to generate a current signal; and a comparison circuit, which is configured to operably compare the current signal with a common mode signal of the current signal, to generate the time point searching result.

12. The switching regulator of claim 8, wherein the time point determination circuit is configured to operably decide whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is the available turn-on time point according to whether the phase node voltage corresponding to each of the occurrence time point is lower than a second voltage threshold.

13. The switching regulator of claim 12, wherein the second voltage threshold includes: a common mode voltage of the phase node voltage.

14. The switching regulator of claim 8, wherein the time point determination circuit is configured to operably count a sequence order of each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result, so as to decide whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is the available turn-on time point.

15. The switching regulator of claim 14, wherein when the sequence order of the occurrence time point of the at least one valley or the at least one peak is an odd, the time point determination circuit decides this occurrence time point of the at least one valley or the at least one peak is the available turn-on time point.

16. The switching regulator of claim 2, wherein the time point deciding unit includes: a frequency divider, which is configured to operably generate a high frequency clock signal and a low frequency clock signal according to the base clock signal, the tolerable frequency range and the PWM signal; and a tolerance period generation circuit, which is configured to operably generate a tolerance period signal according to the high frequency clock signal and the PWM signal, wherein the tolerance period signal is indicative of the tolerance period; wherein when the time point option signal and the tolerance period signal are both at an enable level, the operation clock signal is generated, or otherwise, the operation clock signal is generated at a rising time point of the low frequency clock signal.

17. The switching regulator of claim 16, wherein the time point deciding unit further includes: an edge detection circuit, which is configured to operably detect a rising edge of the low frequency clock signal, to decide the rising time point of the low frequency clock signal.

18. The switching regulator of claim 1, wherein the power stage circuit includes: a buck power stage circuit, a boost power stage circuit, a buck-boost power stage circuit or a flyback power stage circuit.

19. The switching regulator of claim 7, wherein the common mode voltage of the phase node voltage includes: the input voltage.

20. The switching regulator of claim 9, wherein the common signal includes: the input voltage.

21. The switching regulator of claim 1, wherein in a case when the time point option unit is configured to operably generate the lowest voltage time point signal during the tolerance period, the time point option unit includes: a sample-and-hold circuit, which is enabled by a tolerance period signal and which is configured to operably sample and hold a plurality of the phase node voltages during the tolerance period, to generate the time point option signal; and a lowest value selector coupled to the sample-and-hold circuit, wherein the lowest value selector is configured to operably select a lowest phase node voltage among the plurality of the phase node voltages, to generate the lowest voltage time point signal, wherein the lowest voltage time point signal is indicative of the lowest voltage time point.

22. The switching regulator of claim 21, wherein the time point deciding unit includes: a frequency divider, which is configured to operably generate a high frequency clock signal and a low frequency clock signal according to the base clock signal, the tolerable frequency range and the PWM signal; a tolerance period generation circuit, which is configured to operably generate a tolerance period signal according to the high frequency clock signal and the PWM signal, wherein the tolerance period signal is indicative of the tolerance period; and a finite-state machine (FSM), which is configured to operably generate the operation clock signal according to the lowest voltage time point signal and the low frequency clock signal.

23. A control method for controlling a switching regulator, wherein the switching regulator includes: a power stage circuit, which is configured to operably control at least one power switch according to a pulse width modulation (PWM) signal, to switch an inductor coupled to a phase node, thus converting an input voltage to an output voltage; the control method comprising following steps: generating a time point option signal according to a phase node voltage of the phase node during a ringing period subsequent to a blanking period, to indicate at least one available turn-on time point, or generating a lowest voltage time point signal according to the phase node voltage during a tolerance period, to indicate a lowest voltage time point; and deciding the tolerance period according to a base clock signal and a tolerable frequency range and selecting the available turn-on time point or the lowest voltage time point which occurs within the tolerance period to be a decided time point, so as to generate the operation clock signal; wherein the decided time point is a time point which corresponds to a lowest phase node voltage within the tolerance period or an ending time point of the tolerance period; wherein a beginning time point of the tolerance period is a time point obtained by counting a shortest tolerance period from a previous decided time point, wherein the shortest tolerance period corresponds to a highest frequency of the tolerable frequency range; wherein an ending time point of the tolerance period is a time point obtained by counting a longest tolerance period from a previous decided time point, wherein the longest tolerance period corresponds to a lowest frequency of the tolerable frequency range.

24. The control method of claim 23, further comprising: generating a blanking signal which defines the blanking period; wherein a beginning time point of the blanking period is decided according to the PWM signal, and an ending time point of the blanking period is decided according to the phase node voltage, or the ending time point of the blanking period is decided by selecting a time point prior to the beginning time point of the tolerance period.

25. The control method of claim 23, wherein the available turn-on time point is correlated with a

valley time point of a ringing signal of the phase node voltage during the ringing period.

26. The control method of claim 25, wherein the power stage circuit includes a boost power stage circuit, and wherein the output voltage is not lower than 2-fold of the input voltage.

27. The control method of claim 23, wherein the ringing period is correlated with an inductance of the inductor and an equivalent capacitance at the phase node.

28. The control method of claim 24, wherein the step for generating the blanking signal includes: comparing the phase node voltage with a first voltage threshold, to generate a comparison result; and deciding the ending time point of the blanking period according to the comparison result and deciding the beginning time point of the blanking period according to the PWM signal, so as to generate the blanking signal.

29. The control method of claim 28, wherein the first voltage threshold includes: a common mode voltage or a knee voltage of the phase node voltage.

30. The control method of claim 24, wherein the step for generating the time point option signal according to the phase node voltage of the phase node during the ringing period subsequent to the blanking period includes: generating a time point searching result according to the blanking signal and the phase node voltage, wherein the time point searching result indicates an occurrence time point of at least one valley or at least one peak of the phase node voltage within the ringing period; and determining whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is the available turn-on time point, to decide at least one available turn-on time point, thereby generating the time point option signal.

31. The control method of claim 30, wherein the step for generating the time point searching result according to the blanking signal and the phase node voltage includes: executing a differentiation operation on the phase node voltage, to generate a differentiation signal; and comparing the differentiation signal with a common mode signal of the differentiation signal, to generate the time point searching result.

32. The control method of claim 30, wherein the step for generating the time point searching result according to the blanking signal and the phase node voltage includes: executing an integration operation on the phase node voltage, to generate an integration signal; and comparing the integration signal with a common mode signal of the integration signal, to generate the time point searching result.

33. The control method of claim 30, wherein the step for generating the time point searching result according to the blanking signal and the phase node voltage includes: sensing a current flowing through the phase node, to generate a current signal; and comparing the current signal with a common mode signal of the current signal, to generate the time point searching result.

34. The control method of claim 30, wherein the step for determining whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is the available turn-on time point is determined according to whether the phase node voltage corresponding to each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is lower than a second voltage threshold.

35. The control method of claim 34, wherein the second voltage threshold includes: a common mode voltage of the phase node voltage.

36. The control method of claim 30, wherein the step for determining whether each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result is the available turn-on time point includes: counting a sequence order of each of the occurrence time point of the at least one valley or the at least one peak in the time point searching result.

37. The control method of claim 36, wherein when the sequence order of the occurrence time point of the at least one valley or the at least one peak is an odd, deciding this occurrence time point is the available turn-on time point.

38. The control method of claim 24, wherein the step for deciding the tolerance period according to

the base clock signal and the tolerable frequency range and selecting the available turn-on time point or the lowest voltage time point which occurs within the tolerance period to be the decided time point to generate the operation clock signal includes: generating a high frequency clock signal and a low frequency clock signal according to the base clock signal, the tolerable frequency range and the PWM signal; and generating a tolerance period signal according to the high frequency clock signal and the low frequency clock signal, wherein the tolerance period signal is indicative of the tolerance period; wherein when the time point option signal and the tolerance period signal are both at an enable level, the operation clock signal is generated, or otherwise, the operation clock signal is generated at a rising time point of the low frequency clock signal.

39. The control method of claim 38, wherein the step for deciding the tolerance period according to the base clock signal and the tolerable frequency range and selecting the available turn-on time point or the lowest voltage time point which occurs within the tolerance period to be the decided time point to generate the operation clock signal includes: detecting a rising edge of the low frequency clock signal, to decide the rising time point of the low frequency clock signal.

40. The control method of claim 23, wherein the power stage circuit includes: a buck power stage circuit, a boost power stage circuit, a buck-boost power stage circuit or a flyback power stage circuit.

41. The control method of claim 29, wherein the common mode voltage of the phase node voltage includes: the input voltage.

42. The control method of claim 31, wherein the common signal includes: the input voltage.

43. The control method of claim 23, wherein the step for generating the lowest voltage time point signal according to the phase node voltage during the tolerance period includes: When being enabled by a tolerance period signal, sampling and holding a plurality of the phase node voltages during the tolerance period, to generate the time point option signal; and selecting a lowest phase node voltage among the plurality of the phase node voltages, to generate the lowest voltage time point signal, wherein the lowest voltage time point signal is indicative of the lowest voltage time point.

44. The control method of claim 43, wherein the step for deciding the tolerance period according to the base clock signal and the tolerable frequency range and selecting the available turn-on time point or the lowest voltage time point which occurs within the tolerance period to be the decided time point to generate the operation clock signal includes: generating a high frequency clock signal and a low frequency clock signal according to the base clock signal, the tolerable frequency range and the PWM signal; generating a tolerance period signal according to the high frequency clock signal and the low frequency clock signal, wherein the tolerance period signal is indicative of the tolerance period; and generating the operation clock signal according to the lowest voltage time point signal and the low frequency clock signal.
